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Generalized Least Squares Analysis of Stress Levels Based on Sleep Quality, Mental Wellness, Productivity

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Abstract—Work stress is a problem often faced by workers, both employees and freelancers, as it can have a negative impact on their mental well-being and work performance. Several factors are thought to influence stress levels, such as sleep quality, productivity, and mental wellness. This study aims to analyze the influence of these three factors on stress levels, while also evaluating the suitability of the Generalized Least Squares (GLS) method when heteroscedasticity is found in Ordinary Least Square (OLS) regression model analysis. Secondary data consisting of 400 individual records, focusing on respondents who classified as “Employed” and “Self-Employed” were obtained from the “Screen Time vs Mental Wellness Survey-2025.” The initial analysis was performed using a multiple linear regression model with the Ordinary Least Squares (OLS) method. Then, the predictor variables were selected using the backward elimination method until three main predictor variables remained, namely sleep quality, productivity, and mental wellness. After testing, heteroscedasticity was found, so the analysis was continued using the GLS method to obtain more accurate estimation results. The results of this study indicate that sleep quality, productivity, and mental health have a significant effect on stress levels. Individuals with poor sleep quality tend to experience higher stress levels, while individuals with better productivity and better mental wellness generally show lower stress levels. In addition, the application of the GLS model provides more efficient estimation results, as indicated by smaller standard error values compared to the OLS model, without any changes in the significance of the coefficients. Thus, these findings reinforce that the GLS method is a more appropriate approach for analyzing stress data when there are heteroscedasticity assumptions.

Keywords—work stress, sleep quality, productivity, mental wellness, multiple linear regresion, Generalized Least Squares, heteroscedasticity

I. INTRODUCTION

In today's work era portrayed by high pressure and performance demands, work stress has become an increasingly important topic of discourse [1]. According to Beehr and Franz, job stress is an individual's reaction to

demands or pressures emanating from their work environment. This response can occur when individuals perceive that the demands of their work exceed their coping abilities, which can ultimately affect their physical and mental well-being and their performance in the workplace [2]. A 2023 report from The Health and Safety Executive (HSE) also reported 875,000 cases of work-related stress, depression, and anxiety indicating that stress is no longer merely an individual phenomenon but a significant public health issue [3].

In the context of employed and self-employed workers, work stress is often caused by pressure experienced by employees on tasks that cannot be fulfilled. Then, job demands are influenced by workload, time pressure, and personal conflicts experienced by employees in completing their work [4]. Work stress has a significant role in affecting a person's sleep quality. High levels of stress can disrupt sleep patterns and reduce the quality of rest, thus impacting health which in turn can worsen the condition of mental well-being and an individual's ability to manage daily tasks effectively [5].

Sleep quality is a factor closely linked to stress. Several studies have shown that poor sleep quality is strongly correlated with increased stress levels. Research on nurses found that work stress contributes to decreased sleep quality [5] [6]. Similar findings have been reported in lecturers and other formal workers [7] [8]. This suggests that stress arising from workplace pressures can disrupt sleep and reduce the quality of an individual's rest.

In addition to sleep quality, mental health also plays a crucial role in an individual's ability to cope with stress. The better a person's mental health, the lower their stress symptoms [9]. WHO defines mental health as a state of well-being in which an individual is aware of their own abilities, can cope with the normal stresses of life, can work productively, and is able to contribute to their community [10]. Mental health encompasses emotional stability, adaptive coping skills, and psychological balance [11].

Another factor closely related to stress is work productivity. Productivity, especially in the context of full-time work or self-employment, can be affected by various psychological factors, including stress [12]. Although data on productivity and stress is often complex and varied, many studies have found that high levels of stress are associated with decreased productivity [1] [13] [14] [15] [16].

Although numerous studies have linked sleep quality, mental well-being, and stress, most quantitative studies use the Ordinary Least Squares (OLS) approach to estimate this relationship. The OLS estimator gives quite good results when all the classical assumptions in regression are fulfilled, namely the assumptions of normality, homoscedasticity, no multicollinearity, and autocorrelation between the independent variables [17]. Initial diagnostic analysis in this study indicated heteroscedasticity, where the error variance is not uniform across observations. This can lead to inefficient parameter estimation using OLS and biased standard errors.

To overcome this problem, this study uses the Generalized Least Squares (GLS) approach which is designed to handle heteroscedastic error structures [18]. GLS can provide more efficient and reliable parameter estimates than OLS when the homogeneity of variance assumption is not met [18]. Although GLS has been widely used in economics and social sciences, its application in stress research involving workers remains relatively limited. This situation indicates a methodological gap in previous research.

Furthermore, most previous studies have separated the analysis of sleep quality, mental well-being, and productivity, thus failing to examine how these three variables operate simultaneously as predictors of stress levels in a single, statistically adequate model. This indicates a theoretical and empirical gap in the related literature.

Therefore, this study aims to address this gap by implementing a regression model that combines sleep quality, mental well-being, and productivity as predictors of stress, using Generalized Least Squares (GLS) as a more appropriate alternative to OLS. The specific objectives of this study are (1) to determine the direction and weight of the influence of each predictor variable on stress levels in workers and self-employed workers, and (2) to demonstrate the superiority of GLS in providing more reliable estimates when heteroscedasticity is assumed in behavioural data.

II. METHODOLOGY

A. Data Description

The research is based on secondary data obtained from an open-access resource available on the Kaggle Platform, called "Screen Time vs Mental Wellness Survey-2025" [19]. This dataset consists of 400 individual records, providing a comprehensive view of the relationship between digital activity and psychological states. In particular, the data features metrics that outline participants' stress levels, digital work activities, sleep behavior, lifestyle habits, and general mental well-being scores. Only respondents classified as Employed and Self-Employed were included in the analysis. Observations from other occupational categories were excluded to ensure consistency with study objectives. By using this pre-assembled and publicly accessible dataset, which is collected in 2025, the study capitalizes on readily

available, cleaned information to examine factors associated with stress levels among workers.

B. Research Variables

The final set of independent variables was identified using the Backward Elimination technique [20]. This automated procedure began with all potential predictors and systematically eliminated the least impactful variable until only those that significantly contributed to the model's fit were left [20]. As a result, a streamlined final model was created that incorporated just three independent variables: Sleep Quality (X_1), Productivity (X_2) and Mental Wellness-Index (X_3). A summary of these variables can be found in Table 1

Variable	Symbol	Role	Definition and Measurement
Stress Level	Y	Dependent	Participants' self-reported level of stress (Likert Scale, 1-10)
Sleep Quality	X_1	Independent (Dummy)	A binary variable derived from ratings 1, 2, and 3. It indicates participants who reported Low Sleep Quality (rating 1 or 2), using the Moderate Sleep Quality (rating 3) as the baseline reference category.
Productivity	X_2	Independent	Participants' self-assessment of their productivity level (Likert Scale).
Mental Wellness-Index	X_3	Independent	A calculated index representing overall mental well-being (Continuous Scale).

Table 1 Summary of Variables

C. Analysis Procedure

The stages of data analysis conducted to obtain the final multiple linear regression model are organized as follows.

1) Multiple Linear Regression (OLS)

Multiple linear regression is utilized to analyze the relationship between a dependent variable and multiple independent variables. The regression parameters are estimated using the Ordinary Least Squares (OLS) method, which minimizes the sum of squared residuals. Each coefficient represents the expected change in the dependent variable for a one-unit change in the corresponding independent variable, holding other variables constant. This model is commonly used as an empirical approximation when the true functional relationship is unknown and allows the inclusion of categorical variables through dummy coding, provided the model is linear in parameters [21].

In this study, sleep quality is incorporated using dummy variables, with level 3 treated as the reference category. The multiple linear regression is specified as:

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 X_{2i} + \beta_4 X_{3i} + \varepsilon_i \quad (1)$$

The multiple linear regression model is given in (1) where Y_i denotes stress level, D_{1i} and D_{2i} represent sleep quality levels 1 and 2, respectively, X_{2i} is productivity and X_{3i} is the mental wellness index.

2) Outlier and Influence Diagnostics

Prior to conducting diagnostic tests, outlier, and influential observations were examined to ensure the robustness of the regression results. Outliers were identified using standardized residuals, where observations were excluded from the analysis. In addition, leverage values were assessed using the hat matrix. Observations exceeding the threshold of $3\left(\frac{k}{n}\right)$ where k is the number of model parameters and n is the sample size, were removed to mitigate the influence of high-leverage points on parameter estimation.

3) Hypothesis Testing

Hypothesis testing in multiple linear regression is used to evaluate the significance of individual regression coefficients and the overall model. Following standard regression procedures, t-tests were employed to assess partial effects, while the F-test was used to evaluate the joint significance of the explanatory variables

4) Classical Assumption Tests

Following the estimation of multiple linear regression models using the Ordinary Least Squares (OLS) method, classical assumptions tests were conducted to evaluate whether the underlying OLS assumptions were satisfied. These tests include diagnostics for normality, autocorrelation, homoscedasticity, and multicollinearity, and were performed at a 5% significance level.

a) Normality Test:

The normality assumption requires the regression of residuals to be normally distributed. In this study, the Lilliefors test was applied to examine the normality of the residuals. The null hypothesis states that the residuals follow a normal distribution.

b) Autocorrelation Test:

Autocorrelation refers to the correlation among regression residuals across observations. The Durbin-Watson test was employed to detect the presence of first-order autocorrelation. The null hypothesis indicates that there is no autocorrelation in the residuals.

c) Homoscedasticity Test:

The homoscedasticity assumption requires constant variance of regression of residuals. The Breusch-Pagan test was used to assess whether heteroscedasticity is present in the model. The null hypothesis states that the error variance is constant.

d) Multicollinearity Test:

Multicollinearity occurs when explanatory variables are highly correlated, which can distort coefficient estimates. Variance Inflation Factor (VIF) values were computed to evaluate the presence of multicollinearity among the independent variables.

Since the Breusch-Pagan test indicates the presence of heteroscedasticity, the OLS model was re-evaluated

using Generalized Least Squares (GLS) to obtain efficient and reliable parameter estimates.

D. Generalized Least Squares (GLS)

Generalized Least Squares (GLS) is used when OLS assumption of constant error variance is violated, such that $Var(\epsilon) = \sigma^2 V$, where V is a known positive definite matrix [21]. By transforming the model, GLS restores the standard least squares of assumptions and yields efficient parameters of estimates.

To explicitly account for heteroskedasticity, a power of variance function was specified, where the error variance is assumed to depend on the fitted values of the regression model. This variance structure was implemented using a variance-power specification, allowing the GLS estimator to adapt to non-constant error variance across observations.

The GLS estimator is given by:

$$\beta_{GLS} = (X^T V^{-1} X)^{-1} X^T V^{-1} y \quad (2)$$

When V is diagonal, the GLS estimator reduces Weighted Least Squares (WLS) [21].

After estimating the GLS model, the Breusch-Pagan test was reapplied to the normalized residuals to verify whether the heteroskedasticity problem had been adequately addressed.

III. RESULT AND DISCUSSION

A. Data Preparation and Variable Selection

This study utilizes secondary data consisting of various respondent characteristics, including *user_id*, *age*, *gender*, *occupation*, *work_mode*, *screen_time_hours*, *work_screen_hours*, *leisure_screen_hours*, *sleep_hours*, *sleep_quality*, *stress_level*, *productivity*, *exercise_minutes_per_week*, *social_hours_per_week*, and *mental_wellness*. The analysis focuses exclusively on respondents with employment status classified as *employed* and *self-employed*; therefore, observations outside these categories were removed during the initial data preprocessing stage.

	Estimate	Standard Error	T-value	P-value
Intercept	13.2067488	1.0661479	12.387	< 2e-16
Age	0.0018302	0.0071041	0.258	0.796971
Screen Time Hours	0.0505728	0.0387536	1.305	0.193428
Work Screen Hours	-0.0426555	0.0365795	-1.166	0.244989
Leisure Screen Hours	NA	NA	NA	NA
Sleep Hours	-0.0604333	0.0867520	-0.697	0.486865
Sleep Quality 1	-1.7807439	0.3357996	-5.303	3.06e-07
Sleep Quality 2	-1.0497293	0.2712178	-3.870	0.000148
Productivity	-0.0347602	0.0090774	-3.829	0.000173
Exercise Minutes Per Week	0.0016280	0.0008222	1.980	0.049092
Social Hours Per Week	-0.0112766	0.0115408	-0.977	0.329722
Mental Wellness	-0.0858984	0.0079917	-10.748	< 2e-16
Residual Standard Error				0.7625
Multiple R-Squared				0.8592
Adjusted R-Squared				0.852

Table 2 OLS Model Summary (All Variables)

At the exploratory stage, all independent variables were initially included in a multiple linear regression model to

explain variations in stress level. The initial model yielded a coefficient of determination of $R^2 = 85.92\%$, indicating that a substantial proportion of the variability in stress level could be explained by the included predictors. However, retaining all variables in the model may lead to an overly complex and less interpretable model. Therefore, variable selection was conducted using the backward elimination method.

The variable selection process resulted in a final model consisting of three key predictors, namely sleep quality, productivity, and mental wellness. The reduced model achieved a coefficient of determination of $R^2 = 85.62\%$. Although this value is slightly lower than that of the full model, the difference is relatively small, indicating that the reduced model maintains a high explanatory power while using fewer predictors.

B. Ordinary Least Square (OLS) Estimation Results

As a baseline analysis, the relationship between stress level and the selected predictors was first estimated using the Ordinary Least Squares (OLS) method after handling outliers. Sleep quality was incorporated using dummy variables, with good sleep quality (category 3) serving as the reference category. A summary of the ordinary least squares (OLS) estimation method is presented in the table below:

	Estimate	Standard Error	T-value	P-value
Intercept	13.594967	0.435764	31.198	< 2e-16
Sleep Quality 1	-1.769222	0.294576	-6.006	8.77e-09
Sleep Quality 2	-1.013818	0.252615	-4.013	8.45e-05
Productivity	-0.040647	0.008305	-4.895	2.02e-06
Mental Wellness	-0.085425	0.007050	-12.117	< 2e-16
Residual Standard Error				0.7504
Multiple R-Squared				0.8562
Adjusted R-Squared				0.8534

Table 3 OLS Model Summary (All Variables)

The estimated OLS regression model is expressed as:

$$Y_i = 13.595 - 1.769D_{1i} - 1.013D_{2i} - 0.041X_{2i} - 0.0854X_{3i} + \varepsilon_i \quad (3)$$

Where Y_i denotes stress level, D_{1i} and D_{2i} represent sleep quality categories 1 and 2, respectively, X_{2i} denotes productivity, and X_{3i} denotes the mental wellness index.

The intercept value of 13.594967 represents the expected stress level for individuals in the reference category of sleep quality when all continuous predictors are held constant at zero.

The coefficient of Sleep Quality 1 is -1.769222 (p -value = 8.77×10^{-9}), indicating that individuals classified in sleep quality category 1 have an average stress level that is 1.77 units lower than those in the reference sleep quality category, holding other variables constant. Similarly, Sleep Quality 2 has a coefficient of -1.013818 (p -value = 8.45×10^{-5}), suggesting that individuals in this category experience a stress level that is approximately 1.01 units lower than the reference group.

Productivity exhibits a negative and statistically significant association with stress level ($\beta = -0.040647$, p -value = 2.02×10^{-6}), implying that higher productivity is associated with lower stress levels. Likewise, mental wellness shows a negative and significant effect on stress level ($\beta = -0.085425$, p -value < 2×10^{-16}), indicating that

improved mental well-being contributes to a reduction in stress.

C. Diagnostic Tests of OLS Assumptions

To assess the validity of the OLS estimation, several classical assumption tests were conducted. The Lilliefors test (p -value = 0.1062) that the residuals are normally distributed. The Durbin-Watson test indicates no evidence of autocorrelation (p -value = 0.6939), and no multicollinearity is detected, as all Variance Inflation Factor (VIF) values fall below the critical threshold. However, the Breusch-Pagan test yields a p -value of 0.009588, indicating a violation of the homoskedasticity assumption.

From a theoretical perspective, heteroskedasticity implies that the error variance differs across observations. While OLS estimators remain unbiased under heteroskedasticity, they are no longer efficient, and the associated standard errors may be unreliable. Consequently, an alternative estimation approach is required to obtain more efficient parameter estimates.

D. GLS Estimation Results

To address the heteroskedasticity detected in the OLS model, the regression was re-estimated using the Generalized Least Squares (GLS) method with power variance function based on fitted values. The GLS model summary is specified as follows:

	Estimate	Standard Error	T-value	P-value
Intercept	11.442011	0.3171352	30.716	< 2e-16
Sleep Quality 1	-1.769222	0.294576	-6.006	8.77e-09
Sleep Quality 2	0.734081	0.1604268	-4.143	5.04e-05
Productivity	-0.033583	0.0071678	-4.912	1.86e-06
Mental Wellness	-0.082404	0.0076469	-11.838	< 2e-16

Table 4 OLS Model Summary (All Variables)

The GLS estimation results indicate that all predictors remain statistically significant, with coefficient signs largely consistent with the OLS estimates. An one-unit increase in sleep quality is associated with an increase of 0.7341 units in stress level, whereas increases in mental wellness and productivity are associated with decreases in stress level of 0.0824 and 0.0336 units, respectively. These three predictors continue to exhibit meaningful associations with stress level. In particular, productivity and mental wellness retain their negative relationships with stress, indicating robustness across estimation methods.

Furthermore, the standard errors obtained from the GLS model are smaller than those from the OLS model, reflecting improved estimation efficiency. Diagnostic testing on the GLS residuals shows a substantial reduction in heteroskedasticity, confirming that the GLS specification provides a more appropriate model for the data. Therefore, the GLS model is considered the preferred and final specification for explaining stress levels among employed and self-employed individuals.

E. Comparison Between OLS and GLS

Figure 4 presents a direct comparison between the Ordinary Least Squares (OLS) and Generalized Least Squares (GLS) estimates.

	<i>OLS Coef</i>	<i>OLS Standard Error</i>	<i>GLS Coef</i>	<i>GLS Standard Error</i>
Intercept	13.594967	0.435764	11.442011	0.3171352
Sleep Quality 1	-1.769222	0.294576	-1.769222	0.294576
Sleep Quality 2	-1.013818	0.252615	0.734081	0.1604268
Productivity	-0.040647	0.008305	-0.033583	0.0071678
Mental Wellness	-0.085425	0.007050	-0.082404	0.0076469

Table 5 OLS Model Summary (All Variables)

Overall, the coefficient magnitudes and signs remain largely consistent across both models, indicating that the relationships between stress level and its predictors are stable and not sensitive to the estimation method. Importantly, the GLS model produces smaller standard errors for all retained predictors compared to the OLS model. This reduction in standard errors reflects improved estimation efficiency after explicitly accounting for heteroskedasticity in the error structure. Consequently, statistical inference based on the GLS estimates is more reliable than that obtained from the OLS model.

Although minor differences in coefficient magnitudes are observed, particularly for the sleep quality categories, these coefficients should be interpreted relative to the reference category (sleep quality level 3). The overall direction and significance of the predictors remain consistent, reinforcing the robustness of the estimated relationships.

The findings of this study highlight the importance of sleep quality, productivity, and mental wellness in explaining stress levels among employed and self-employed individuals. Across both OLS and GLS estimations, these variables consistently exhibit statistically significant relationships with stress, indicating robustness of the results.

Sleep quality shows a significant association with stress level, where individuals with poorer sleep quality experience different stress outcomes compared to those with good sleep quality. This result aligns with existing literature suggesting that sleep plays a critical role in emotional regulation and stress recovery. Insufficient or poor-quality sleep can impair cognitive functioning and increase vulnerability to stressors, particularly in working populations.

Productivity is found to be negatively associated with stress level, implying that individuals who perceive themselves as more productive tend to experience lower stress. This finding supports theories of work-related stress which argue that a sense of task accomplishment and efficiency can reduce psychological strain. Higher productivity may also reflect better time management and coping strategies, which mitigate stress.

Mental wellness exhibits the strongest negative association with stress level among all predictors. This result is consistent with psychological models that position mental well-being as a key protective factor against stress. Individuals with higher mental wellness are likely to possess better emotional resilience, enabling them to manage work-related and social stressors more effectively.

The presence of heteroskedasticity in the OLS model suggests that the variability of stress levels differs across individuals, potentially due to unobserved heterogeneity in personal or occupational characteristics. By applying the GLS method, this study accounts for such heteroskedasticity and obtains more efficient estimates. The similarity in

coefficient signs and magnitudes between OLS and GLS further confirms the stability of the estimated relationships.

Overall, these findings emphasize that interventions aimed at improving sleep quality, enhancing productivity, and promoting mental wellness may be effective strategies for reducing stress among working individuals. However, this study is subject to limitations, including the use of secondary data and self-reported measures, which may introduce measurement bias. Future research could incorporate longitudinal data or objective health indicators to further validate these relationships.

IV. CONCLUSION

This study aims to analyze how sleep quality, productivity, and mental wellness affect stress levels in both employed and self-employed individual. It also looks at whether the Generalized Least Squares (GLS) method is a suitable way to analyze data when there are issues with heteroskedasticity. Based on the results, both objectives of the study have been successfully achieved.

The results reveal that all three predictors have a significant effect on stress levels. People who sleep poorly tend to feel more stressed, which shows the importance of good rest to manage psychological pressure well. Productivity shows a negative relationship with stress, suggesting that higher perceived productivity contributes to lower stress levels. Of all the factors examined, mental wellness has the greatest impact, showing that maintaining good mental health is especially important for handling stress at work.

The analysis also found that the usual Ordinary Least Squares (OLS) method doesn't satisfy one of its important assumptions, which could affect the estimator inefficient, and the associated standard error may be unreliable. By switching to the Generalized Least Squares (GLS) method, this problem was solved, resulting in more efficient parameter estimates with smaller standard errors while maintaining consistent coefficient signs and significance levels. This shows that GLS is a better and more reliable way to analyze stress-related data when the error variance differs across observations (heteroskedasticity).

V. ACKNOWLEDGMENT

The authors would like to thank Bina Nusantara University for the facilities and academic environment throughout this research. Special appreciation is extended to Margaretha Ohyver, S.Si., M.Si., for her guidance and feedback during the completion of this study.

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